Olasılık teorisi, Rastgele Değişkenler ve Rastgele Prosesler:

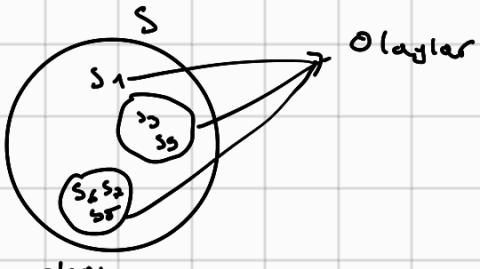
Deney (experiement): zar atışı

Sonuç (outcome)

Örnek uzayı (sample space) : $S = \{1, 2, 3, 4, 5, 6\}$

Olay (event) : $A = \{\text{çift sayılar}\} = \{2, 4, 6\}$

$A \subset S$



$$P(A) = \sum_{\forall s_i \in A} P(s_i) = P(2) + P(4) + P(6)$$

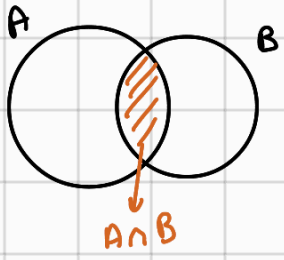
$$= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

Eşit olasılıklı sonuçlar için $P(A) = \frac{|A|}{|S|} = \frac{\text{Anın eleman sayısı}}{\text{Sınır}} = \frac{3}{6} = \frac{1}{2}$

Bir olayın olması için olaydaki tek bir sonucun olması yeterli.

Kesin olay (Certain event) $= S \rightarrow P(S) = 1$

İmkansız olay (Impossible event) $= \emptyset \rightarrow P(\emptyset) = 0$



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$A \cap B = \emptyset \Rightarrow$ A ve B kesişimsiz olaylar

(mutually exclusive, disjoint)

$$A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B) - \underbrace{P(A \cap B)}_{0}$$

$$P(A \cup B) = P(A) + P(B)$$

$A' = \{a \mid a \in S, a \notin A\}$ uzayın A dışındaki tümelemanlar

$A \cup A' = S$

$A \cap A' = \emptyset$

$S = A \cup A'$

$P(S) = P(A \cup A') = P(A) + P(A') - P(A \cap A')$

$P(A \cap A') = \emptyset \rightarrow 0$

$1 = P(A) + P(A')$

$\Rightarrow P(A') = 1 - P(A)$

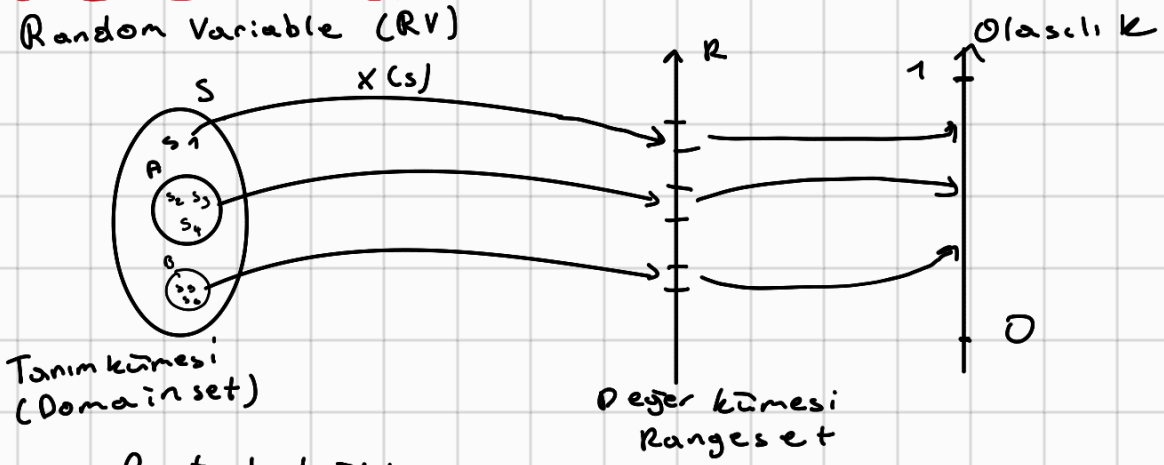
Bağımsızsa (Independent)

$A \perp\!\!\!\perp B \Leftrightarrow P(A \cap B) = P(A) \cdot P(B)$

Rastgele Değişken (RD)

Random Variable (RV)

$X(s)$: RD



Rastgele değişken

$X(s) = X$

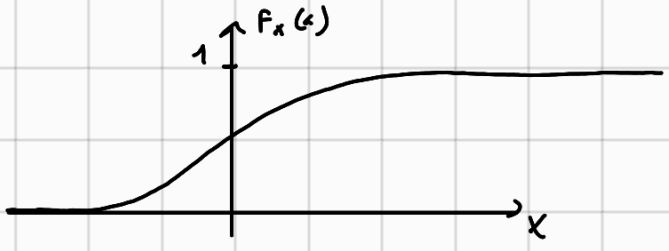
x : X RD'nin belli bir değeri.

örnek Y, y

Olasılık Dağılım Fonksiyonu (ODF)

Cumulative Distrubution function - CDF

$F_X(x) = P(X \leq x)$ (olasılık)



- $F_x(-\infty) = 0$
- $F_x(+\infty) = 1$
- $0 \leq F_x(x) \leq 1$
- $x_1 \leq x_2 \Rightarrow F_x(x_1) \leq F_x(x_2)$

Azalmayan fonksiyon (Nondecreasing function)

Olasılık Yoğunluk Fonk. (OYF)

Probability Density Function (PDF)

Bir x rastgele değişkenin belli bir bölgede olmasının olasılığını altında kalan alanla gösteren fonksiyona denk

$f_x(x)$ OYF

$$P(X \in A) = \int_A f_X(x)\,dx \qquad \text{OYF}$$

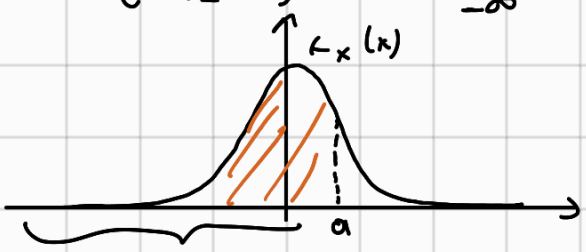
$$P(X \le x) = \int_{-\infty}^{x} f_X(t) = F_X(x) \qquad \text{ODF}$$

$$f_X(x) = \frac{d}{dx} F_X(x)$$

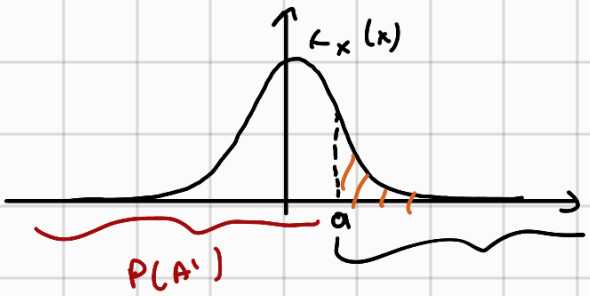
- $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$
- $\forall x \; f_X(x) \ge 0$

azalmayan fonksiyonun türevi pozitiftir

1) $P(X \le a) = \int_{-\infty}^{a} f_X(u)\,du = F_X(a)$

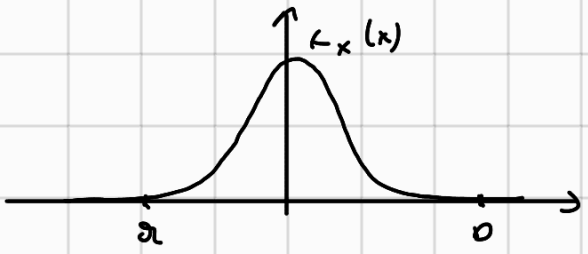


2) $P(\underbrace{X > a}_{A}) = \int_{a}^{\infty} f_X(u)\,du = 1 - \underbrace{\int_{-\infty}^{a} f_X(u)\,du}_{P(A')} = 1 - F_X(a)$



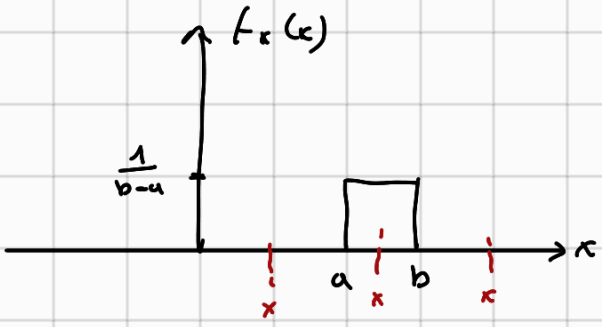
3) $P(a < X \le b) = \int_{a}^{b} f_X(u)\,du = \int_{-\infty}^{b} f_X(u)\,du - \int_{-\infty}^{a} f_X(u)\,du$

$= F_X(b) - F_X(a)$



Düzgün Dağılım (Uniform distribution)

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{diğer} \end{cases}$$



$$F_X(x) = \int_{-\infty}^{x} f_x(u)\,du$$

$x < a$ (α)

1. bölge : $x < 0 \Rightarrow f_X(x) = 0 \Rightarrow F_x(x) = 0$

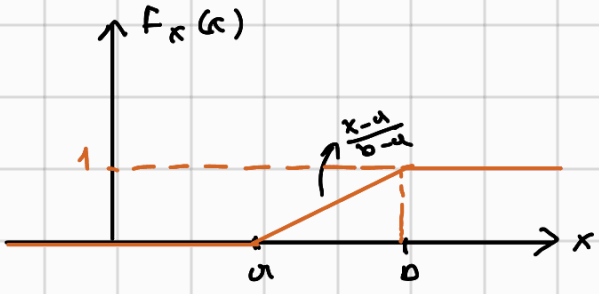
2. bölge : $a < x < b \Rightarrow F_x(x) = \int_{-\infty}^{x} f_x(u)\,du = \int_a^x \frac{1}{b-a}\,du = \frac{u}{b-a}\Big|_a^x$

$$= \frac{x-a}{b-a}$$

3. bölge : $x > b \Rightarrow F_x(x) = \int_{-\infty}^{x} f_x(u)\,du = \int_a^b \frac{1}{b-a}\,du = \frac{u}{b-a}\Big|_a^b$

$$= \frac{b-a}{b-a} = 1$$

$$F_x(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a < x < b \\ 1, & x \geq b \end{cases}$$



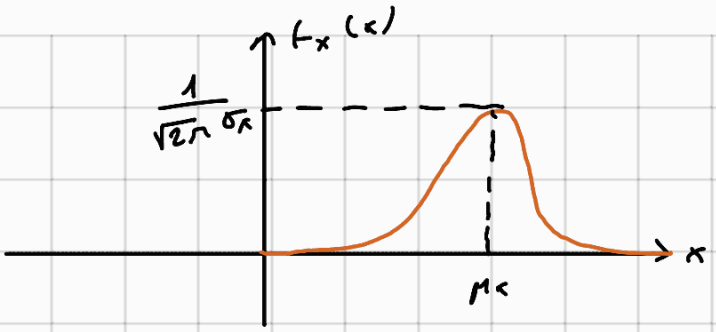
Normal Dağılım (Gauss)

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma_x} e^{-\frac{(x-\mu_x)^2}{2\sigma_x^2}} = N(\mu_x, \sigma_x^2)$$

μ_x : ortalamadeğer (mean)

σ_x : Standartsapma

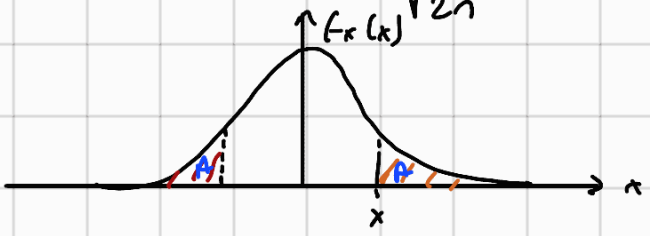
σ_x^2 = Varyans



$$F_X(x) = \int_{-\infty}^{x} f_X(u)\,du = \frac{1}{\sqrt{2\pi}\,\sigma_X} \int_{-\infty}^{x} e^{\frac{-(u-\mu_X)^2}{2\sigma_X^2}}\,du$$

Standart Gauss Dağılımı

$$N(0,1): \quad f_X(x) = \frac{1}{\sqrt{2\pi}}\, e^{-\frac{x^2}{2}}$$



$$Q(x) = P(X > x) = \int_{x}^{\infty} f_X(t)\,dt = \frac{1}{\sqrt{2\pi}} \int_{x}^{\infty} e^{-\frac{t^2}{2}}\,dt \quad , \; x \geq 0$$

S 104 S Table B1 $x \geq 0$ Q tablosu

$Q(0) = 0.5$

$Q(-x) = 1 - Q(x)$

$$\frac{u - \mu_X}{\sigma_X} = t$$

$$u = t\sigma_X + \mu_X \qquad du = \sigma_X\,dt$$

$u = -\infty \rightarrow t = -\infty$

$u = x \rightarrow t = \frac{x - \mu_X}{\sigma_X}$

$$F_X(x) = \int_{-\infty}^{x} f_X(u)\,du = \frac{1}{\sqrt{2\pi}\,\sigma_X} \int_{-\infty}^{x} e^{\frac{-(u-\mu_X)^2}{2\sigma_X^2}}\,du = \frac{1}{\sqrt{2\pi}\,\sigma_X} \int_{-\infty}^{\frac{x-\mu_X}{\sigma_X}} e^{-\frac{t^2}{2}}\,\sigma_X\,dt$$

$$= 1 - \frac{1}{\sqrt{2\pi}} \int_{\frac{x-\mu_X}{\sigma_X}}^{\infty} e^{-t^2/2}\,dt = 1 - Q\left(\frac{x-\mu_X}{\sigma_X}\right)$$

$Q\left(\frac{x-\mu_X}{\sigma_X}\right)$

Beklenen Değer (Expectation)

$$\mu_x = E[X] = \int_{-\infty}^{+\infty} x\, f_X(x)\, dx$$

↳ X'in beklenen değeri (ortalaması)

$$E[X^n] = \int_{-\infty}^{+\infty} x^n f_X(x)\, dx$$

X'in n. momenti

$$E[g(x)] = \int_{-\infty}^{+\infty} g(x) f_X(x)\, dx$$

$$g(X) = (X-\mu_x)^2 \Rightarrow \sigma_x^2 = E[(X-\mu_x)^2] = \int_{-\infty}^{\infty} (x-\mu_x)^2 f_X(x)\, dx$$

$$= \underbrace{\int x^2 f_X(x)\, dx}_{E[X^2]} + \mu_x^2 \underbrace{\int f_X(x)\, dx}_{1} - 2\mu \underbrace{\int_{-\infty}^{\infty} x f_X(x)}_{\mu_x}$$

$$\sigma_x^2 = E[X^2] - \mu_x^2 = E[X^2] - (E[X])^2$$

Karakteristik fonk

$$\Phi_X(w) = E[e^{jwx}] = \int_{-\infty}^{+\infty} e^{jwx} f_X(x)\, dx$$

$$\boxed{E[X^n] = (-j)^n \cdot \frac{d^n \Phi_X(w)}{dw^n}\Big|_{w=0}}$$

e^{wx} olduğunda moment üreten fonksiyon deniyor

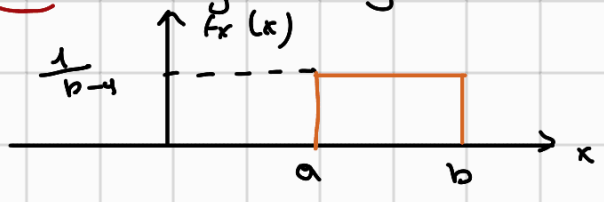
Son zamanlarda

Moment üreten fonk (MÜF): $M_X(s) = E[e^{sX}] = \int_{-\infty}^{\infty} e^{sx} f_X(x)\, dx$

↳ laplace dönüşümü

$$E[X^n] = \frac{d^n}{ds^n} M_X(s)\Big|_{s=0}$$

Örnek: Düzgün dağılım



$$\mu_x = E[X] = \int_{-\infty}^{\infty} x f_x(x)\,dx$$

$$= \int_a^b x \frac{1}{b-a}\,dx = \frac{1}{b-a}\frac{x^2}{2}\Big|_a^b$$

$$= \frac{b^2-a^2}{2(b-a)} = \frac{a+b}{2} \quad //$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_x(x)\,dx = \int_a^b x^2 \frac{1}{b-a}\,dx = \frac{1}{b-a}\frac{x^3}{3}\Big|_a^b$$

$$= \frac{b^3-a^3}{3(b-a)} = \frac{a^2+ab+b^2}{3} \quad //$$

$$\sigma_x^2 = E[X^2] - \mu_x^2 = \frac{a^2+ab+b^2}{3} - \left(\frac{a+b}{2}\right)^2$$

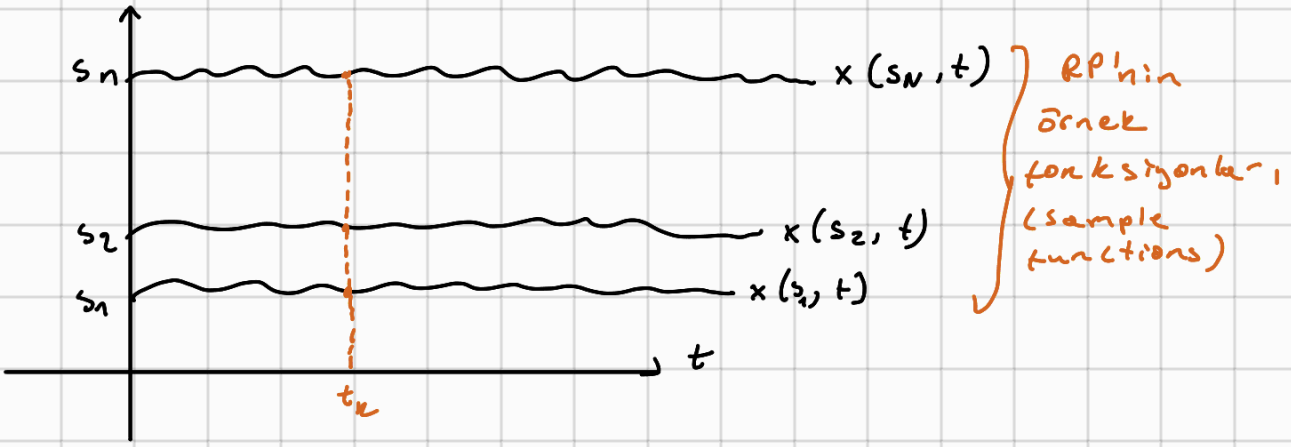
$$= \underbrace{\frac{a^2+ab+b^2}{3}}_{(4)} - \underbrace{\frac{a^2+b^2+2ab}{4}}_{(3)}$$

$$= \frac{a^2-2ab+b^2}{12} = \frac{(b-a)^2}{12}$$

$$\mu_x = \frac{a+b}{2}$$

$$\sigma_x^2 = \frac{(b-a)^2}{12}$$

Rastgele Prosesler (Random, stochastic)



RP : $X(s,t) \equiv X(t)$

↳ Örnek fonksiyonlar topluluğu.

$X(t_k) = X_{t_k} = X_k$ RD

$$E[X_{tk}] = \int_{-\infty}^{\infty} x_{tk} \cdot f_{X_{tk}}(x_{tk})\, dx_{tk} = \mu_{X_{tk}}$$ RP'in beklenen değeri.

RP'in özilişki (auto correlation) fonk

$$R(t_1, t_2) = E[\underbrace{X(t_1)\, X(t_2)}_{X_{t_1} \cdot X_{t_2}}] = \int_{-\infty}^{\infty}\int x_{t_1} \cdot x_{t_2}\, \underbrace{f_X(x_{t_1}, x_{t_2})}_{x_{t_1} \text{ ve } x_{t_2}}\, dx_{t_1}\, dx_{t_2}$$

RP'lerinin ortak OYF (joint PDF)

Stasyonerlik

- **Sıkı Stasyoner (SS)** : Bütün momentler zamana bağlı değil

  Strict sense stationary (SSS)

- **Gevşek Stasyoner (GS)** : ilk 2 moment zamana bağlı değil

  Wide sense stationary (WSS)

Bir RP aşağıdaki şartları sağlıyorsa GS'dir

1) $E[X(t)] = \mu_x = \text{sabit}$
2) $R_x(t_1, t_2) = R_x(\underbrace{t_1 - t_2}_{\tau}) = R_x(\tau) = E[X(t)\, X(t-\tau)]$

   2. moment $\tau = t_1 - t_2$ ye bağlı

Process SS $\Rightarrow$ GS

$\not\Leftarrow$

$\Leftarrow$ Gauss RP sağlar

Haberleşme sistemlerinde GS önemli

Özellikle

- $R_x(-\tau) = R_x(\tau)$ çift fonk.
- $R_x(0) \geq |R_x(\tau)|$ → complex sayılar için
- $R_x(0) = E[|X(t)|^2]$ Güç

Rastgele Processler Güç işaretleridir burda istatistiksel ortalama alınıyor

- $R_X(\tau) \overset{F}{\longleftrightarrow} S_X(f)$ GYF
güç yoğunluk konksiyon

GYF

RP Güç işareti → GYF, $S_X(f)$

- $R_X(\tau) \overset{F}{\longleftrightarrow} S_X(f)$ $\frac{\text{watt}}{\text{hertz}}$
- $S_X(f) \geq 0$ (Reel değerli)

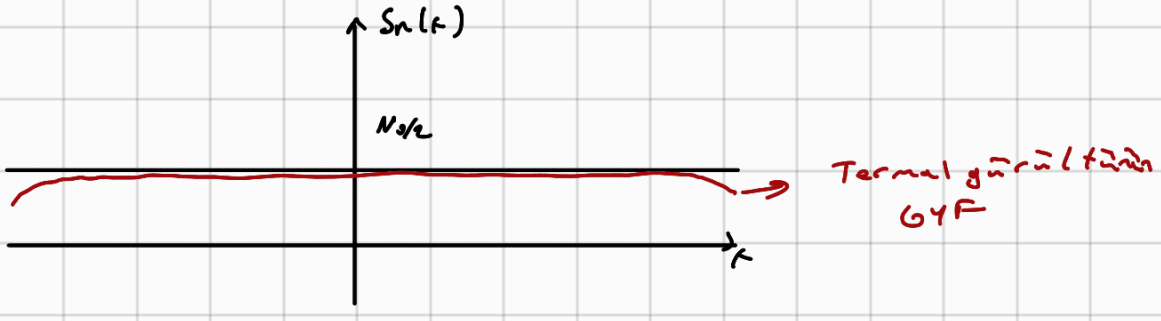
- Prosesin ortalama gücü

$$P_X = R_X(0) = \int_{-\infty}^{+\infty} S_X(f)\, df$$

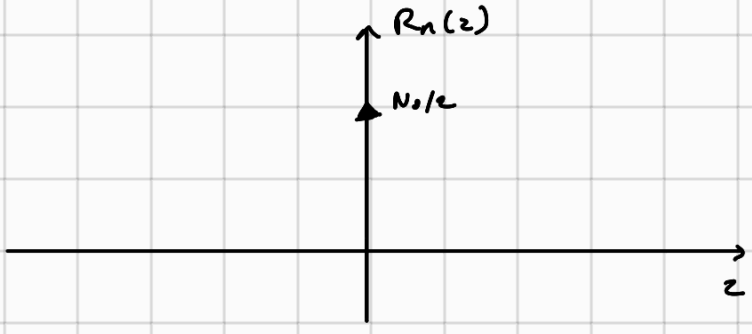
- Reel değerli $X(t)$ RP için $S_X(-f) = S_X(f)$

Beyaz Gürültü RP
(White Noise RP)

$n(t) \rightarrow S_n(f) = \frac{N_0}{2} = \text{sabit}$ $\frac{\text{Watt}}{\text{Hz}}$ $\forall f$



$$R_n(\tau) = \frac{N_0}{2}\delta(\tau)$$

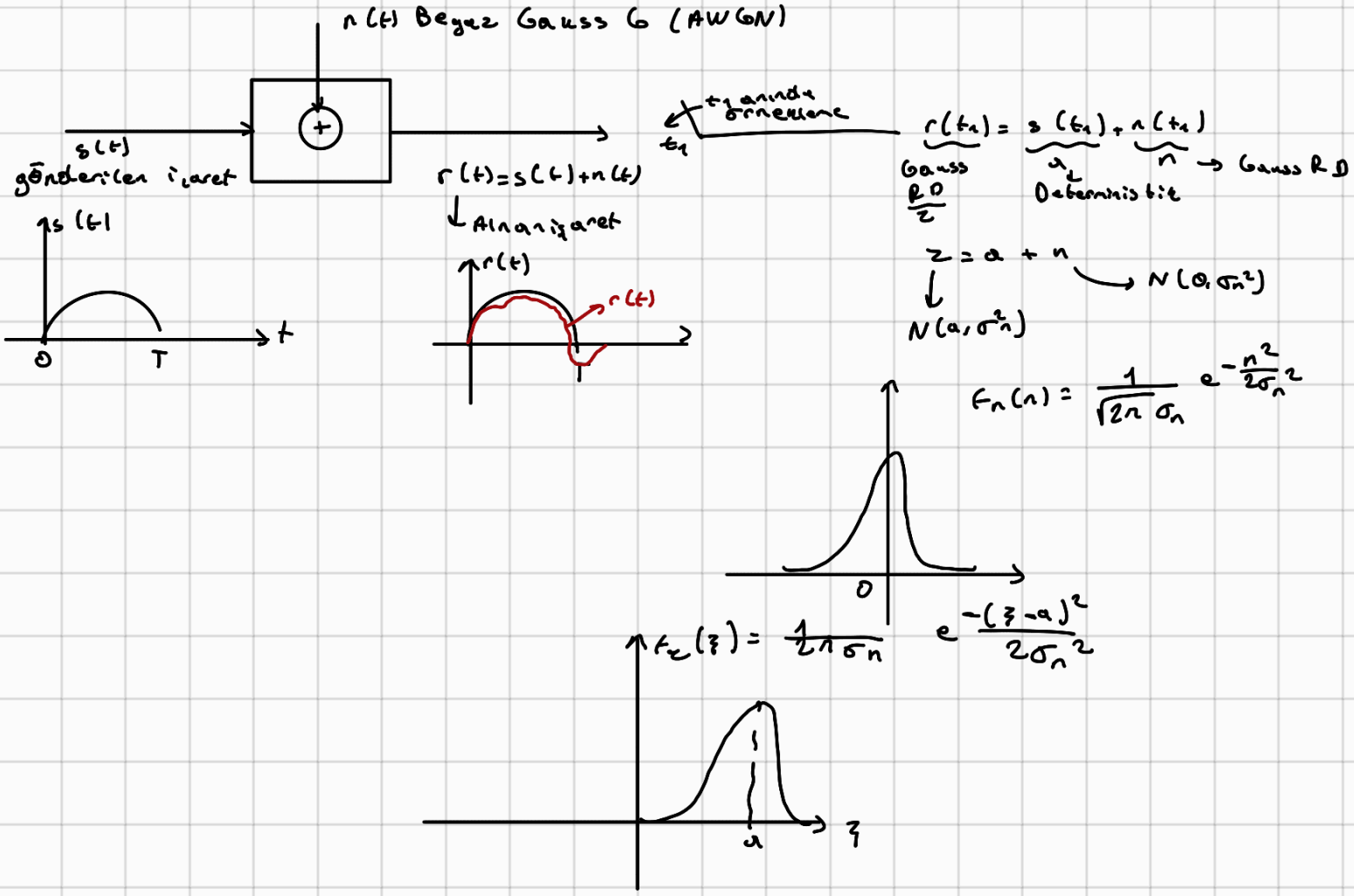


Gücü sonsuz

Gerçekte gücü sonsuz güce sahip işaret yok

Bütün frekanslarda olduğu için Beyaz gürültü denir.

(Prizma)



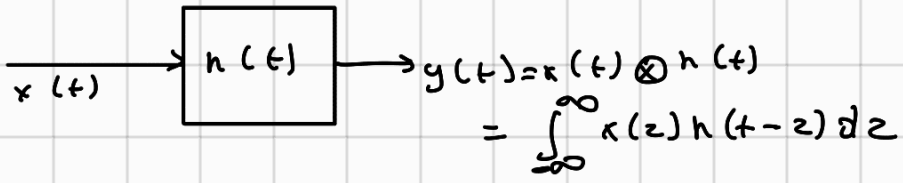
Alıcı alırken z pozitif mi negatif mi diye bakıyor

s(t) pozitifken z(t)'nin negatif olma olasılığım

" negatifken " pozitif " "

bulup toplayıp ikiye bölerek yapıyor

LZD sistemler



X(t) RP

$S_X(f)$

Y(t) RP

$S_Y(f) = S_x(f) \, |H(f)|^2$

İşaretler ve Sistemler

(-T/2 , T/2) arasında harcanan enerji

$$E_x^T = \int_{-T/2}^{T/2} |x(t)|^2 dt$$

güç $P_x{}^T = \frac{E_x{}^T}{T}$

$$E_x = \lim_{T\to\infty} E_x{}^T = \lim_{T\to\infty} \int_{-T/2}^{T/2} |x(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$P_x = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

$E_x < \infty \Rightarrow x(t)$ enerji işareti:

$P_x < \infty \Rightarrow x$ güç işareti:

$x(t)$ periyodik $\Rightarrow P_x = \frac{1}{T_0} \int_{T_0} |x(t)|^2 dt$

$$x(t) = \begin{cases} A\cos(2\pi f_0 t) \\ A\sin(2\pi f_0 t) \end{cases} \Rightarrow P_x = \frac{A^2}{2} \text{ watt}$$